

Speculative Future Imagined Through Satellite Imageries with Generative AI

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Abstract

"Water City" visualizes a speculative city in 2300 affected by sea level rise, using generative AI. The semi-fictional satellite images depict coastal scenes and urban landscapes, stimulating viewers' imagination and helping them perceive climate change as a personal issue. Balancing scientific accuracy and artistic spontaneity, this approach fosters dialogue about speculative futures. By blending fiction and reality, it encourages viewers to imagine the future of their own city, turning abstract environmental issues into concrete, personal challenges. This method demonstrates the potential of image generation AI for emotional data visualization and creative discussions.

CCS Concepts

• **Applied computing** → **Media arts**; • **Human-centered computing** → **Visualization theory, concepts and paradigms**.

Keywords

Generative AI, Satellite Imagery, Science Communication, Speculative Design, Future Visualization,

ACM Reference Format:

Yuki Yoshida and Akira Wakita. 2024. Speculative Future Imagined Through Satellite Imageries with Generative AI. In *SIGGRAPH Asia 2024 Art Papers (SA Art Papers '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3680530.3695454>

1 INTRODUCTION

The impact of climate change is an urgent global concern. While scientists work to predict and mitigate these effects, artists explore speculative methods to visualize and communicate potential futures. Recent advancements in generative AI are expanding its potential in science communication, making global issues feel more personal by creating immersive and emotionally engaging visual narratives.

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SA Art Papers '24, December 03–06, 2024, Tokyo, Japan

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ACM ISBN 979-8-4007-1133-6/24/12
<https://doi.org/10.1145/3680530.3695454>

2 ABOUT "Water City" PROJECT

2.1 Overview of the work

"Water City" visualizes semi-fictional satellite imagery of cities in 2300, depicting a scenario where sea levels have risen due to climate change. The work illustrates a 5.4-meter sea level rise using image-generative AI and topographic data, resulting in a realistic yet fictional satellite image.

2.2 Scientific data basis

The scientific basis for sea level rise comes from the "Fifth Assessment Report" of the Intergovernmental Panel on Climate Change (IPCC), which outlines socio-economic scenarios based on different greenhouse gas concentration pathways, known as Representative Concentration Pathways (RCP). In the most controlled scenario, sea levels are projected to rise by 0.6 to 1.07 meters by 2300, while in the worst-case scenario, a sea level rise of 2.4 to 5.4 meters is predicted [IPCC 2019]. This work visualizes the worst-case scenario, using the extreme value to create a more visually dramatic effect and to highlight the experimental nature of the visualization method.

2.3 Contrasting with hazard simulations

Extensive efforts have been made to simulate sea level rise and visualize inundation areas, such as the "Earth Apps Sea Level Rise Simulator" [JAXA 2024], which uses ALOS topographic data to simulate flooded areas in blue based on user-selected water levels.

Simulators and hazard maps typically focus on emergency disaster communication. However, sea level rise due to climate change gradually alters cityscapes over time. "Water City" explores how cities may evolve with rising water levels, examining both the negative impacts of environmental change and potential adaptations in urban life, fostering dialogue rather than emphasizing dystopian risks.

3 METHODS

3.1 Satellite imagery and inundation data preparation

To prepare the satellite data and inundation area data, Google Earth Engine (GEE) [Google 2024] was utilized. GEE is an online platform that combines satellite images with a computing engine, enabling efficient processing and analysis of geospatial data.

The base optical satellite images used were "Harmonized Sentinel-2 MSI Level-2A" [ESA 2023] provided by the European Space Agency (ESA) under the Copernicus program. These images offer atmospherically corrected 10-meter ground resolution of the Earth's

surface with simple cloud removal processing was applied on GEE. The image was synthesized to represent the average from May to October 2023, as the region is covered in snow during the winter.

For the simulation of flooded areas, the elevation data “MERIT DEM (Multi-Error-Removed Improved-Terrain Digital Elevation Model)” [Yamazaki et al. 2019] was used. MERIT DEM combines multiple satellite observation data and proprietary terrain analysis algorithms to provide high-precision elevation data covering the entire globe, with a resolution of approximately 90 meters. Using this elevation data, areas with elevations below 5.4 meters were identified.

3.2 AI image generation using Stable Diffusion

The next step involves processing the images with an image generation AI, Stable Diffusion [CompVis 2022], an open-source image generation AI based on a latent diffusion model [Rombach et al. 2022]. Stable Diffusion also allows fine-tuning through LoRA (Low-Rank Adaptation) [Hu et al. 2021], enabling additional training of pre-trained models. For this project, 34 characteristic images from Sentinel-2 images of the Sapporo region were extracted by dividing them into 512 x 512 pixel tiles, and a Sapporo-specific LoRA model was created.

The image generation process was performed in the following steps (Figure 3):

- (1) Prepare the original satellite image.
- (2) Create a mask for inundated areas using GEE.
- (3) Refine the mask by aligning its boundaries with the actual terrain and smoothing rough edges.
- (4) Combine the original image and the refined mask, then apply LoRA-based Image-to-Image processing to generate the output image.

The generated images show realistic depictions based on additional training data, such as developed ports at the boundary with the sea.

During image generation, unnatural areas were manually corrected using image processing software, followed by inpainting to fine-tune the images. This method effectively adjusted realistic transportation like bridges and roads. Due to machine power limitations, the images were divided into tiles and stitching them together at the end.

4 AS A VIDEO WORK

4.1 Exhibit in a public space

“Water City” was exhibited as part of a collaborative video installation at Sapporo International Art Festival. From February 4, 2024, it was displayed for a week on a 14.4 meter wide and 2.7 meter high LED screen in a commercial facility, attracting approximately 1,200 visitors on the first day. (Figure 1)

4.2 “Water City” as a video work

As a video installation in a public space, part of the video featured a “Tourist guideboard of 2300”, introducing fictional tourist spots with AI-generated local mascots, common in Japanese tourist sites (Figure 4).



Figure 1: The exhibition space at the art festival

Visitors at the actual location feel a sense of discrepancy when gathering information, realizing it’s slightly different from reality. They soon discover the display depicts their current place in the year 2300. Semi-fictional information intervening into everyday life provides an opportunity to overlay and contemplate the distant future with their present.

4.3 Viewer reactions

At the exhibition, a notebook was provided where viewers could freely write their thoughts about the collaborative video installation (Figure 2). Among the entries related to “Water City”, the following examples were observed:

“A map of Sapporo in the year 2300. I felt a sense of sadness, not from the idea of something suddenly breaking, but rather from the slow, gradual end of things. Although I won’t be alive by then, I still felt this sadness.”

“The future is frightening. Will it really be happy?”

“I love Taiwan Tangerine juice, so I’m looking forward to the Bay Area.”

“(In response to the question ‘What will be gone 100 years from now?’ written in the note) Winter sports will be gone.”

Viewers expressed both positive and negative emotions, such as “sadness” and “looking forward. They imagined everyday and concrete future scenarios, including elements from the work like Taiwan Tangerine juice, and others beyond it, like winter sports. This suggests semi-fictional data visualization effectively stimulated their imagination and connection to daily life.

Table 1: A Framework for Expression and Cognition in Data Visualization

	Rational Cognition Cognition based on objective data and logic. Visualization encourages a universal interpretation.	Emotional Cognition Cognition based on individual emotions and intuition. Visualization promotes personal interpretations driven by imagination and empathy.
Abstract Expression	An expression that simplifies data and information into basic forms to facilitate logical understanding. Examples: Graph, Chart, Diagram.	An expression that uses abstract shapes and patterns to stimulate aesthetic perception and evoke intuitive reactions. Examples: Generative Art, Motion Graphics
Realistic Expression	An expression that faithfully reproduces reality to deepen scientific and technical understanding. It seeks accurate representation based on real-world information. Examples: Geospatial data visualization, Medical imaging	An expression that evokes emotional responses through realistic visuals and storytelling, raising specific issues. Examples: Sci-Fi films, Documentary photography

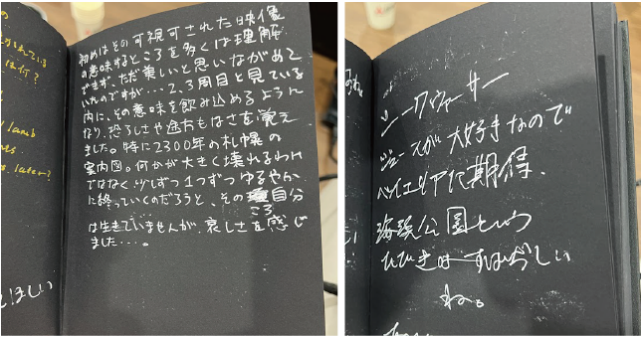


Figure 2: The notebook provided at the exhibition

5 DISCUSSION

5.1 Emotional engagement through blending fiction and reality

The detailed semi-fictional scenes created by AI made it possible to stimulate the viewers’ imagination. The project showed that image generation AI effectively simulates inundated areas and their surroundings. The images realistically depict the boundary between water and land, including coastal scenes like ports. Larger coastal buildings mirror real-life patterns from areas with factories and warehouses. These photorealistic images do more than show inundation; they suggest that “people live here,” allowing viewers to envision a future version of their own city after sea level rise. The goal is to make climate change feel personal through a blend of fiction and reality.

This approach aligns with Speculative Design by blending fiction and everyday life, encouraging the exploration of new possibilities beyond the constraints of reality [Dunne and Raby 2013]. By combining the real and the imagined, the discrepancy between reality and fiction is used to explore future possibilities while reflecting on the present. This helps engage viewers emotionally, fostering awareness and potentially influencing their behavior toward environmental issues. If we adhere to strict scientific accuracy, by the year 2300, not only the sea level but many aspects of the cityscape will have changed. However, to stimulate the viewer’s imagination,

it is important to depict a future that overlaps with present everyday life, rather than a completely transformed world. A similar approach can be seen in notable AI projects such as “Deep Empathy” project [MIT Media Lab and UNICEF 2017], which uses AI to visualize familiar urban landscapes affected by disasters, and Mila’s “This Climate Does Not Exist” [Mila 2021], described as “an AI-driven experience based on empathy, allowing users to imagine the environmental impacts of the current climate crisis, one address at a time.”

Furthermore, in this project, the use of a “fictional tourist map” allowed viewers to connect even more deeply with familiar everyday contexts. The speculative questions posed on the map encouraged viewers to reflect on their own imagined future, and based on viewer reactions, it can be said that this technique worked effectively to inspire speculative thinking about the future.

5.2 Realistic expression and emotional cognition in data visualization

This study develops a framework categorizing two expression types in data visualization and their cognitive effects (Table 1).

The approach in “Water City” falls into the “Realistic Expression and Emotional Cognition” category. Realistic visuals create an emotional impact by accentuating the contrast with the viewer’s everyday life, thereby reinforcing the intended worldview. Achieving this blend traditionally required advanced editing techniques or costly production for photorealistic computer graphics. However, AI-based image generation now allows individuals to create “realistic yet fictional” imagery more affordably. “Water City” showcases how this innovation can bring new possibilities to data visualization.

6 CONCLUSION

This report examines the effectiveness of image generation AI as an emotional data visualization technique, using “Water City” as a case study to illustrate the fusion of realistic expression and emotional cognition in science communication. Generative AI enables the creation of realistic yet fictional imagery that stimulates viewers’ imagination. By employing this technology, the project demonstrates the potential for new forms of data visualization that not only enhance imagination but also foster emotional engagement

and dialogue. This approach is expected to contribute to the development of visualization methods that encourage deeper discussions about the future.

Acknowledgments

I would like to express my sincere gratitude to Mitsubishi Electric Corporation Integrated Design Center for their collaboration in this research and the collaborative video installation. I also extend my appreciation to the members of Akira Wakita Laboratory for their support and contributions, as well as to my family for their encouragement.

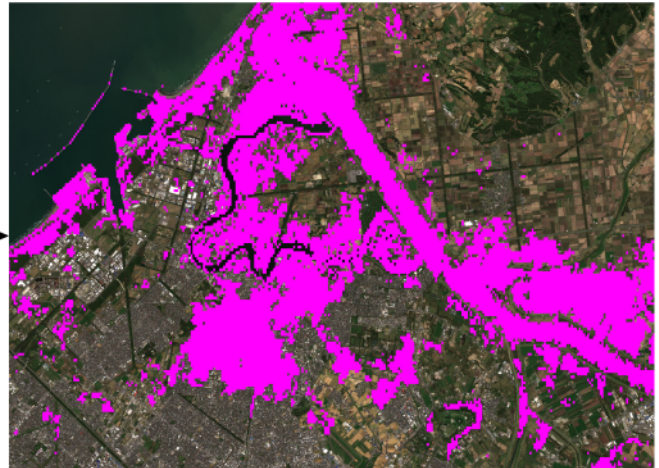
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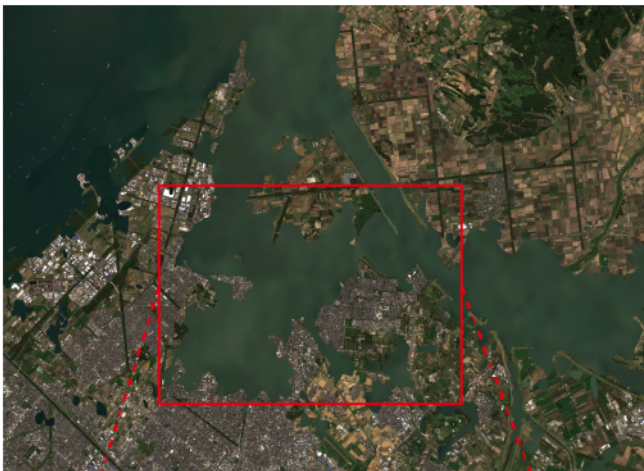
1. Original



2. Mask for inundated areas



3. Refine the mask



4. Image-to-Image AI generation



Input (detailed)



Output (detailed)



contains modified Copernicus Sentinel Data (2023)

Figure 3: Image Generation Process. While the input is simply filled with sea color, the generated image depicts a coastline that is developed like a real city.



Figure 4: The “Tourist guideboard of 2300” displayed in the video sequence includes the following features: 1) A list of fictional tourist spots. 2) A mascot character provides ‘Hints for imagination’ for each spot. 3) The section shows a climate advanced by global warming.